

KWEKU-ABEIKU ATTAH ANYEN

+233 550-076-503 | kaa.anyen@gmail.com | linkedin.com/in/kwekuanyen/

PROFILE SUMMARY

Results-oriented professional specializing in Software Engineering and Artificial Intelligence with understanding of secure system design. Strong foundation in computer vision, NLP, and deep learning with hands-on experience in CNNs, RNNs, LSTMs, and vision-language models. Proficient in full-stack development using React, Node.js, TypeScript, and AWS. Trained in Agile methodologies (Scrum, Kanban) and applying security frameworks (SABSA, NIST, ISO 27001) to software development lifecycle.

EDUCATION

Ashesi University

Master of Philosophy - Intelligent Computing Systems

Berekuso, Eastern Region

Expected Graduation: May 2027

Focus: Artificial Intelligence, Machine Learning, Computer Vision, Natural Language Processing, Software Architecture

Relevant Courses: Intelligent Computing, Computer Vision, NLP, Deep Learning, Software Architecture, Secure Software Development, Agile Project Management

Ashesi University

Bachelor of Science - Computer Science

Berekuso, Eastern Region

Graduated: May 2021

Relevant Courses: Data Structures, Algorithms, Web Development, Mobile Development, Software Engineering

CERTIFICATIONS/AWARDS

Agile Project Management: From Principles to Practice - McGill University

In Progress

PMI Authorized Training Partner certification - 35 Professional Development Units (PDUs). Comprehensive coverage of Agile Scrum, Lean, Kanban, Extreme Programming, planning, estimation, risk management, team coaching, and retrospectives.

PUBLICATIONS

P. Mireku, **K.-A. Attah-Anyen**, A. Nartey, N. Nanka-Bruce, and B. Blankson. "Do Safety-Aligned Vision-Language Models Degrade Differently Under Common Image Corruptions?" *Workshop on Generative Models for Computer Vision, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2026.

generative-vision.github.io/workshop-CVPR-26/papers/21.pdf

- First controlled base-vs-aligned evaluation of vision-language models under the ImageNet-C corruption suite across four matched model pairs and three alignment paradigms (MPO, instruction tuning, SFT+RLHF)
- Identified and corrected a verbosity-scoring mismatch that biases standard VQA exact-match metrics against aligned models
- Showed alignment paradigm is a stronger predictor of robustness change than model scale, with a severity-dependent reversal under defocus blur

WORK EXPERIENCE

Synero Labs – Accra, Ghana

June 2026 - Present

Software Engineering Intern — Full-Stack & Platform (Boafo)

- **Boafo Runtime** (Python, FastAPI, LangGraph multi-tenant agent engine): built at-rest Fernet encryption and a tenant-scoped secret store so tool configs hold opaque refs and cross-tenant resolution is rejected
- Wired post-generation guardrail verdicts into the live LangGraph loop so hallucination, scope, and unconfirmed-action failures halt or escalate a turn instead of reaching the user; added ~75 tests across the routing chain
- Hardened DeepSeek and Claude adapters for OpenAI-compatible gateways: configurable base URLs and model IDs, classifier-tier override, and handling of 200-with-error payloads so retry and fallback actually fire
- **Boafo App**: built the insurance case workspace (case list, six-panel workspace, docked chat, inbox, approval and reversal flow) in Next.js 16 / React 19, and modelled industry verticals as a config object so Insurance and OMC share one component set
- Moved the case domain off fixtures onto PostgreSQL (Prisma schema, shared Zod types, NestJS module, tRPC router) and added a platform-staff grant for the internal admin console without weakening Membership-scoped access to customer data
- Built typed assistant response cards (generic table, keyword resolver, per-kind components) and shipped the public marketing site as a separate Next.js app so it deploys independently and cannot import credential-bearing server code
- **Boafo Commerce**: shipped RBAC with team invites and clerk/owner rules, Organization/Branch/Membership tenancy, encrypted local and cloud backups with restore, and an internal support console (feature flags, merchant impersonation, business suspension)

- Led UI/UX discovery and redesign direction for Bofo ERP; designed the config-driven “Project Workspace” pattern reused across Insurance, Logistics, and OMC verticals

iSpace Foundation – Accra, Ghana

January 2025 - August 2025

Video and Visual Department Lead

- Developed engaging social media content (videos, graphics, photos) supporting marketing and program promotion
- Shot and edited videos and documentaries during foundation events using Adobe Creative Suite
- Conducted research on emerging trends in African tech ecosystem
- Measured and reported on social media campaign impact

Wode Maya – Remote

February 2025 - August 2025

Video Editor, Documentarian & Videographer

- Edit high-quality video content for Africa’s most influential YouTuber
- Apply advanced post-production techniques including color correction, sound mixing, motion design
- Collaborate on storyboarding and visual direction ensuring clarity and consistency
- Optimize content formats for YouTube and digital platforms

iSpace Foundation – Accra, Ghana

March 2021 - December 2024

Media Assistant

- Developed social media content supporting company programs
- Shot and edited video and photography during foundation events
- Assisted in measuring social media campaign impact

Brothers in Hue – Ghana

August 2019 - Present

Co-Founder & Media Specialist

- Co-founded creative media agency offering photography, videography, drone media, motion design
- Expanded services during COVID-19 to include graphic design meeting client demand
- Introduced website development services enabling full digital solutions
- Evolved offerings to include 3D animation, VFX, AR experiences, advanced motion graphics
- Managed client relationships, project timelines, creative direction maintaining high delivery standards

AirtelTigo – Ghana

June 2018 - August 2018

Tech Support

- Managed and maintained user accounts on AirtelTigo platform
- Implemented account security measures including provisioning, deactivation, reactivation
- Monitored user activity for potential security threats and anomalies

PROJECTS AND RESEARCH

AI & Machine Learning Projects

CRESCENDO - AI-Powered Project Management

April 2026

- Built AI-powered project management application for Opus 4.7 Hackathon
- Developed Board-State Awareness Engine injecting complete project context into AI calls
- Built full-stack application: React 19, TypeScript, Node.js, Fastify, PostgreSQL
- Implemented RBAC with 5 roles, JWT session management, comprehensive audit logging
- Designed 746-line build document covering 10 development sprints

U-Net Medical Image Segmentation

April 2026

- Reimplemented U-Net architecture in PyTorch for biomedical image segmentation
- Evaluated performance across 4 medical imaging datasets (EM, ultrasound, CT, MRI)
- Achieved Dice scores: 0.864 (ISBI), 0.864 (BUSI), 0.971 (Spleen), 0.909 (Hippocampus)
- Conducted ablation study: skip connections critical for small structures (tumor Dice -15.9% without)

Vision-Language Model Corruption Robustness

May 2026

- Evaluated corruption robustness of aligned VLMs (InternVL, Qwen2-VL, Gemma)
- Tested across GQA, VQAv2, TextVQA with 4 corruption types at 3 severity levels
- Found MPO alignment reduced corruption errors by 17% (Relative mCE: 0.830)
- Created reproducible evaluation pipeline with strict experimental constraints

Automatic Speech Recognition Implementation

2025-2026

- Developed ASR system using LibriSpeech dataset (1000+ hours of speech)
- Implemented and benchmarked: vanilla RNNs, attention-based RNNs, GRUs, bidirectional LSTMs
- Audio preprocessing: MFCC and spectrogram feature extraction

- Evaluated using Word Error Rate (WER) and Character Error Rate (CER) metrics

MediVoice - Healthcare AI Application

2025-2026

- Developed healthcare application for clinical consultation transcription
- Implemented ASR pipeline for real-time doctor-patient conversation transcription
- Built NER system for medical entity extraction (symptoms, diagnoses, medications)
- Applied medical domain adaptation for specialized vocabulary handling

Image Classification & Computer Vision with CNNs

2025-2026

- Designed and implemented custom CNN architectures for image classification
- Implemented batch normalization, dropout regularization, data augmentation
- Applied transfer learning with pre-trained models for improved performance

Software Engineering & Systems Projects

Boafo Ticketing - Event Ticketing Platform (Sole Engineer)

2026

- Built a Turborepo with a NestJS public API, Next.js organizer dashboard, and embeddable Next.js checkout (Prisma, PostgreSQL, Redis, BullMQ) letting organizers sell from their own site
- Enforced correctness in code rather than review: single-writer capacity, server-side price computation, integer-pesewa money, and mandatory Idempotency-Key on writes — with architectural tests that fail CI when the rules break
- Shipped Paystack payments (card and mobile money), HMAC-signed QR tickets, an offline-capable phone scanner, refunds, transfers, and a sold-out waitlist
- Published the API as a product: generated OpenAPI spec, secret vs. publishable keys, and signed webhooks for payment, check-in, and refunds
- Load-tested an on-sale (800 concurrent buyers, 100 tickets, zero oversell) and caught a reverse-proxy rate-limit misconfiguration that would have rejected real buyers after ~8 requests
- Deployed as a Docker Compose stack (Caddy TLS, three apps, Postgres, Redis, one-shot migrator) on a single droplet, with the same images runnable locally

EcoDrone - Environmental Monitoring System

2025-2026

- Collaborated on drone-based environmental monitoring with aerial data collection
- Developed autonomous flight logic using Parrot Olympe SDK
- Integrated temperature and CO2 sensors with Raspberry Pi companion computer
- Managed flight operations: Pixhawk 6C controllers, QGroundControl, real-time telemetry

Wave - University Delivery Application

2025-2026

- Built university delivery application streamlining campus logistics
- Managed full-cycle project planning: requirements, timeline, resource allocation
- Created UI/UX designs for delivery personnel and customer interfaces
- Drafted comprehensive master plan documentation and technical specifications

Pillr - Organization Management System

2025-2026

- Engineered management system for organizational structure and member tracking
- Designed scalable multi-tenant architecture supporting multiple organizations
- Implemented user invitation workflows with RBAC and hierarchical permissions
- Managed domain configuration: custom domain setup, DNS, SSL certificate provisioning

Security Architecture & Secure Software Development

Secure SDLC Architecture (mPharma Case Study)

May 2026

- Designed secure software development lifecycle for pharmaceutical platform (850+ pharmacies, 9 countries)
- Conducted STRIDE threat modeling across system components (POS, e-commerce, inventory, payments)
- Defined security testing strategy: SAST, DAST, dependency scanning, penetration testing
- Addressed supply chain security: WordPress plugin backdoors, Axios npm compromise
- Applied NIST SSDF, OWASP ASVS, ISO/IEC 27034 standards

Integrated Risk Management Strategy

May 2026

- Developed risk management strategy for African-European expansion (Ghana/Nigeria to UK/Germany)
- Applied STRIDE analysis, DREAD scoring, ALE calculations for ransomware, data breaches, GDPR fines
- Conducted GDPR gap analysis with country-by-country compliance comparison
- Control selection with ROI analysis: \$280K investment → \$793K ALE reduction (283% ROI)

- Applied SABSA framework across all 6 layers (Contextual, Conceptual, Logical, Physical, Component, Operational)
- Studied NIST Cybersecurity Framework, ISO 27001, GDPR for practical application
- Developed security metrics: MTTD, MTTR, patch compliance, vulnerability density

TECHNICAL SKILLS

Programming Languages: Python, JavaScript/TypeScript, HTML, CSS, Java, C#, C++, SQL, Go

AI/ML & Research: PyTorch, TensorFlow, Keras, NumPy, Pandas, Matplotlib, scikit-learn, Computer Vision (CNNs, U-Net), NLP (RNNs, LSTMs, GRUs, Transformers), Speech Processing (ASR), Vision-Language Models, LangGraph, LLM Agent Safety (guardrails, HITL, fail-closed routing)

Software Development: React, Next.js, Node.js, NestJS, Fastify, Express, FastAPI, tRPC, Prisma, PostgreSQL, MySQL, Redis, BullMQ, REST APIs, OpenAPI, GraphQL, Turborepo, Docker, Caddy, AWS (ECS, RDS, S3), Paystack integration

Security & Architecture: SABSA, STRIDE, DREAD, NIST (SSDF, RMF, CSF), ISO 27001, OWASP ASVS, Threat Modeling, Risk Assessment, Secure SDLC, Multi-Tenant Isolation, Credential-at-Rest Design, RBAC, Architectural Testing

Creative Technology: Adobe Creative Suite (Photoshop, Illustrator, Premiere Pro, After Effects), Motion Graphics, VFX, 3D Animation, AR Content Creation, Drone Media

Robotics & IoT: Raspberry Pi, Pixhawk flight controllers, Parrot Olympe SDK, QGroundControl, sensor integration, autonomous systems

Other: Git, GitHub, CI/CD, Agile Methodologies (Scrum, Kanban, Sprint Planning, Retrospectives), SDLC, Project Management, Technical Writing, Team Leadership

REFERENCES

Available upon request